PRATEEK VASHISHTHA

+91-932-9654-621 • Noida, India • prateekvashishtha6@gmail.com • LinkedIn • GitHub • LeetCode

SUMMARY

Final year CSE student at Vellore Institute of Technology with hands-on experience in backend development and machine learning through an internship at ICAR-IISWC, where I worked on real environmental data pipelines. I enjoy building things end-to-end — from writing clean Python backends to connecting them with usable frontends. Most of my projects sit at the intersection of data and software, and I like keeping code modular enough that it can actually be maintained.

EDUCATION

- **Bachelor of Technology in Computer Science and Engineering** (Sep 2023 – Sep 2027)

Vellore Institute of Technology, India
Cumulative Grade Point – **8.70 / 10.0**
Courses: Data Structures, Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning

WORK EXPERIENCE

Indian Council of Agricultural Research – IISWC (Dec 2025)
SDE Intern Dehradun, India (On-site)

Backend and Data Workflow Engineering for Soil Conservation Platform

- Redesigned and modularized legacy data processing workflows in Python, improving code maintainability and reducing processing time for soil conservation analysis pipelines by introducing structured, reusable components.
- Upgraded the ML analytical pipeline by integrating additional environmental feature sets using Pandas and NumPy, improving the accuracy and depth of soil erosion risk predictions and supporting better field decision-making.
- Refactored the automation layer for preprocessing and reporting workflows, reducing manual intervention and enabling end-to-end execution from raw dataset ingestion to final report generation with minimal configuration.
- Enhanced the FastAPI backend by introducing RESTful endpoints for real-time data querying and analytical reporting, improving the platform's responsiveness and usability for researchers accessing environmental datasets.

PROJECTS

Krishi Rakshak *[Live]* *[GitHub]*

- Built a full-stack machine learning platform for soil conservation analysis, ingesting multi-source agricultural datasets and applying predictive models to assess soil erosion risk at a field level, enabling data-driven conservation planning.
- Engineered modular Python preprocessing pipelines that automated data validation, feature extraction, and normalization, reducing manual data preparation effort and ensuring consistent inputs to downstream ML models.
- Designed a FastAPI backend exposing RESTful endpoints for real-time inference and historical trend queries, integrated with a Next.js and Tailwind CSS frontend for interactive dashboards accessible to field researchers.
- Implemented SQL-based data persistence and query workflows for storing prediction results and environmental readings, supporting longitudinal analysis across multiple agricultural zones.
- *Tech: Python, FastAPI, Pandas, NumPy, Machine Learning, SQL, Next.js, Tailwind CSS, REST APIs* Dec 2025

Employee Attrition Prediction Management System *[Live]* *[GitHub]*

- Developed an end-to-end predictive analytics system to identify employee attrition risk patterns from structured HR datasets, covering the full pipeline from raw ingestion through model deployment and reporting.
- Performed extensive data cleaning, preprocessing, and feature engineering on multi-dimensional HR records, handling missing values, encoding categorical variables, and selecting high-signal features to improve model interpretability.

- Trained and evaluated XGBoost and Scikit-learn classification models with cross-validation and hyperparameter tuning, achieving high precision in identifying at-risk employees and surfacing actionable retention signals.
- Integrated SQL-based data workflows for efficient batch analytics and connected Matplotlib visualizations to generate executive-ready reports highlighting attrition trends by department, tenure, and role.
- *Tech: Python, Pandas, NumPy, SQL, XGBoost, Scikit-learn, Matplotlib* Nov 2025

TrustIQ | AI Verification Platform *[GitHub]*

- Developing a backend-first AI platform for structured data validation and intelligent document verification, architecting API-driven ingestion pipelines that route incoming documents through automated analysis and integrity checks.
- Designed a modular backend using FastAPI and PostgreSQL with clearly separated service layers for document parsing, LLM-assisted analysis, and result storage, enabling independent scaling and testing of each component.
- Integrated Transformer-based NLP models for document understanding and field extraction, reducing manual verification overhead and improving throughput for high-volume document processing workflows.
- Building a Next.js and Tailwind CSS interface for real-time verification status tracking, providing structured audit trails and confidence scores for each document processed through the platform.
- *Tech: Python, FastAPI, SQL, Transformers, REST APIs, Next.js, Tailwind CSS, PostgreSQL* Ongoing

SKILLS

- **Production biased:** Python, FastAPI, REST APIs, SQL, Machine Learning, Predictive Analytics, Backend Architecture
- **Proficient:** Pandas, NumPy, XGBoost, Scikit-learn, Data Cleaning, Exploratory Data Analysis, OOP
- **Implementation:** PostgreSQL, Docker, AWS, Transformers, LLM Pipelines, Data Analysis, Feature Engineering
- **Familiar:** Java, C++, Kubernetes, Postman, Matplotlib, Seaborn, Cloud AI, Data Scraping
- **Fundamentals:** Data Structures, Algorithms, System Design (HLD), DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

- AWS Certified Developer - Associate June 2026
- AWS Certified AI Practitioner June 2026
- Oracle Certified Data Science Professional Sep 2025
- Deloitte Data Analytics Job Simulation Jul 2025
- Introduction to Machine Learning, NPTEL Sep 2024

POSITIONS OF RESPONSIBILITY

- **Lead, Public Relations and Outreach**, UI/UX Club, VIT Bhopal (Aug 2025 – Mar 2026)
  Spearheaded outreach strategy for the UI/UX Club, designing and executing multi-channel campaigns that expanded the club's visibility across campus and drove a measurable increase in event attendance and membership applications.
  Coordinated end-to-end logistics for workshops, design sprints, and inter-club collaborations, ensuring smooth execution and strengthening cross-functional ties with technical and non-technical student communities.
- **Core Member, Content Team**, AI Club, VIT Bhopal (Nov 2024 – Mar 2025)
  Produced technical articles, explainers, and workshop materials on machine learning and AI topics, distilling complex concepts into accessible content that engaged students across all experience levels.
  Contributed to planning and running AI-focused events including speaker sessions and hands-on ML workshops, helping grow an active learning community around artificial intelligence at VIT Bhopal.